

macOS Homework Setup Guide

VS Code + Node.js LTS + global TypeScript

Goal: Install VS Code, Node.js LTS, and TypeScript once, then compile and run a local TypeScript file.

Official Links

- Visual Studio Code: <https://code.visualstudio.com/docs/setup/mac>
- Node.js: <https://nodejs.org/en/download/>
- TypeScript: <https://www.typescriptlang.org/download/>

Before You Start

- Download only from the official links in this guide.
- Never say, display, screenshot, or send the computer administrator password.
- If a step is blocked, save the exact error and bring it to the teaching team or Hangout Session.

Need setup help?

Join the assistant-teacher Hangout Session from 4:30-5:30 PM Pacific, as scheduled in the Microsoft Teams Hangout Session channel. Bring this guide, the step number, and the exact error message.

Step 1 - Install Visual Studio Code

- 1 Open the official macOS setup link above.
- 2 Download Visual Studio Code for macOS. The Universal build supports both Apple silicon and Intel-based Macs.
- 3 Open the downloaded .dmg file.
- 4 Drag Visual Studio Code.app into Applications.
- 5 Open VS Code from Applications and confirm that the Welcome screen appears.

Checkpoint

VS Code opens from Applications and shows the Welcome screen.

Step 2 - Install Node.js LTS

- 1 Open the official Node.js download link above.
- 2 Choose the version marked LTS. Do not choose Current.
- 3 Download and open the normal macOS installer.
- 4 Keep the normal choices and finish the installation.

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- 5 Close every VS Code window, then reopen VS Code.

Node.js runs JavaScript. Its installer also provides npm, which installs TypeScript.

Step 3 - Open the VS Code Terminal

Choose Terminal > New Terminal. The normal macOS shell should say zsh.

Run one command at a time:

```
node --version  
npm --version
```

Both commands should print a version number.

If a command is not found

Close every VS Code window, reopen it, start a new terminal, and try again. If it still fails, save the exact error for Hangout.

Step 4 - Install TypeScript Globally

Try these commands first:

```
npm install -g typescript  
tsc --version
```

The tsc command should print a TypeScript version. This global beginner setup makes tsc available in future lesson folders without another TypeScript installation.

If macOS reports EACCES or a permission error

Stop and save the exact error. Do not repeatedly change permissions. Use `sudo npm install -g typescript` only when the teaching team directs you. Never say, display, screenshot, or send the computer administrator password. If unsure, bring the error to Hangout.

Create and Run local-ready.ts

- 1 In VS Code, choose File > Open Folder.
- 2 Create or choose KidsComputerScience.
- 3 Create 2026-08-01-homework-setup inside it and open that folder.
- 4 If Workspace Trust appears, trust the folder only because you created it.

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- 5 Create local-ready.ts and type the program below.

```
const score: number = 82;

if (score >= 90) {
  console.log("Legendary");
} else if (score >= 70) {
  console.log("Hero");
} else {
  console.log("Keep training");
}
```

Save the file, then run these commands one at a time:

```
tsc local-ready.ts
node local-ready.js
```

Expected result

The compiler creates local-ready.js. Node.js runs it, and the terminal prints Hero. Change score to 69, compile again, and confirm that Keep training prints.

macOS Troubleshooting

- node or npm is not found: confirm the Node.js LTS installer finished, close VS Code, and reopen it.
- The global install reports a permission error: stop, record the exact error, and use the teaching-team help path above.
- tsc is not found: confirm the global install finished, open a new terminal, and run tsc --version again.
- tsc local-ready.ts prints nothing: that can mean it worked. Look for local-ready.js in the Explorer.
- Node cannot find local-ready.js: confirm the terminal is inside the homework folder and that local-ready.js exists.

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Final Green Checks

- ☐ VS Code opens from Applications.
- ☐ The VS Code terminal uses zsh.
- ☐ `node --version` works.
- ☐ `npm --version` works.
- ☐ `tsc --version` works.
- ☐ `local-ready.ts` compiles into `local-ready.js`.
- ☐ `node local-ready.js` prints a message.

Submission reminder

Complete the attached homework PDF and submit it to both Microsoft Teams and Ishwari Raut ma'am. If a box remains unchecked, bring the exact error to Hangout.